VAISHNAVI K

Electronics and Communication Engineering Graduate | VLSI | Verilog

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with 8.5/10 CGPA and hands-on learning experience in VLSI, Design for Testability (DFT), Verilog and Digital Electronics. Experienced in developing and simulating a 4-bit scan chain using Verilog with a testbench and waveform analysis. Strong interest in semiconductor technology, digital IC design and testing. Quick learner with a strong foundation in electronics and programming, seeking an opportunity to contribute and grow as a Young Graduate Trainee at Infineon Technologies.

EDUCATION

B.E. Electronics and Communication Engineering -AVS Engineering College
2022-2026 | CGPA: 8.5/10

HSC – Government Girls Higher Secondary School, Salem
2021-2022 | 83%

SSLC – Government High School, VedugathamPatti, Salem
2019-2020 | 83%

TECHNICAL SKILLS

- VLSI & Digital Design: Design for Testability (DFT), Scan Chain, Digital Logic Design, Flip-Flops, Shift Registers
- HDL: Verilog
- Programming: C, Java
- Simulation & Development: Icarus Verilog, Circuit Verse,Vivado, VS Code, Eclipse
- Digital Electronics: Logic Gates, Adders, Subtractors, Multiplexers, Digital Circuits
- Other**:** HTML, CSS, MS Word, Excel, PowerPoint

VLSI PROJECT

Design for Testability (DFT) – 4-Bit Scan Chain Using Verilog

- Designed a 4-bit scan chain using Verilog to improve digital circuit testability.
- Implemented scan-enable control for normal parallel operation and scan/shift operation.
- Designed serial data shifting using scan-in and scan-out paths.
- Developed a Verilog testbench to verify functional and scan operations.
- Simulated the design and analyzed waveforms to verify circuit behavior.
- Gained practical understanding of DFT concepts, scan testing, RTL coding and simulation.

Tools/Technologies: Verilog | DFT | Scan Chain | Digital Logic | RTL Simulation

INTERNSHIP

VLSI Internship – TechVedhu Academy

- Studied fundamentals of **VLSI design, digital circuits and Design for Testability**.
- Worked on a DFT-based scan-chain project using Verilog.
- Practiced RTL coding, testbench development and simulation.
- Analyzed simulation waveforms to verify the design.

Advanced Embedded Systems Intern — Caliber Embedded Technologies, Salem

- Gained hands-on exposure to embedded system architecture and microcontroller programming.
- Worked with sensors, real-time systems and hardware interfacing concepts.
- Learned fundamentals of Embedded C and hardware simulation.

ACADEMIC PROJECT

AI Driven Adaptive Power Management System for Hybrid Power Plants with BESS

- Developed a hybrid renewable-energy system integrating solar, wind and BESS.

- Implemented AI-based control concepts for efficient power distribution.
- Used Arduino to monitor voltage, current and load conditions.
- Integrated ThingSpeak IoT for real-time monitoring.
- Focused on power stability and battery protection.

CERTIFICATIONS

- C Programming
- Java Programming
- HTML and CSS
- VLSI

CORE STRENGTHS

Problem Solving | Analytical Thinking | Quick Learning | Teamwork | Adaptability